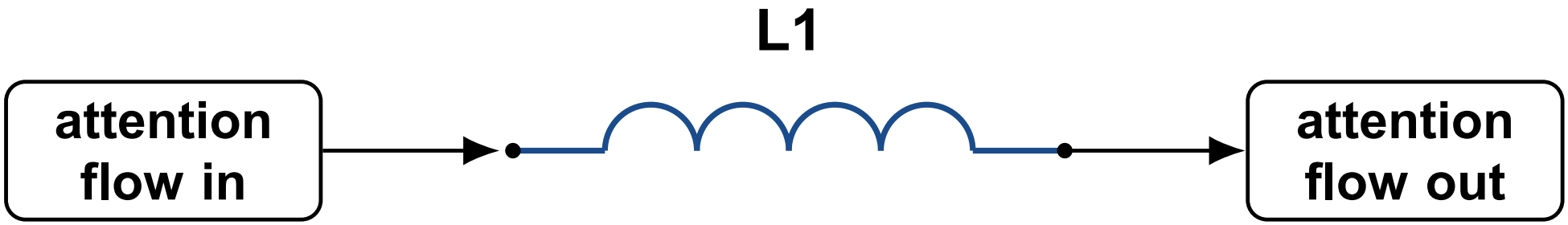
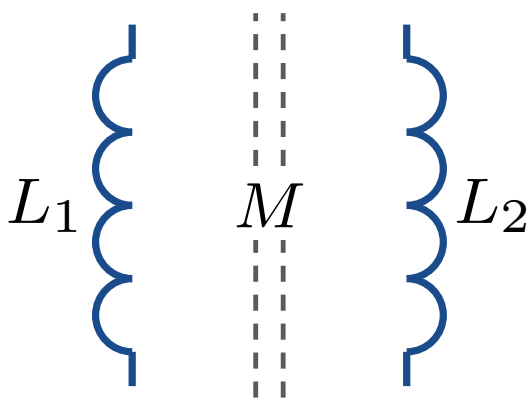
(a) COGNITIVE INDUCTOR L1



inductance L in h·Ω_c: $L = k_d \times H_{\text{practice}}$
opposes the *rate*, not the flow: $V_{\text{opp}} = L\,|di/dt|$
depth coefficient k_d = 0.5 / 1.0 / 2.0 / 4.0 (O / E / C / P)

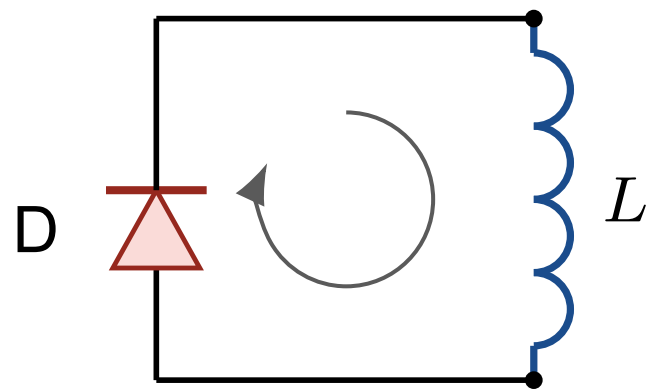
(b) MUTUAL INDUCTANCE



$$M = k\sqrt{L_1 L_2}$$

k = coupling, $0 \leq k < 1$

(c) FLYBACK DIODE



blocks while current flows;
conducts on field collapse